

# NO CONSTANT IN THE SPHERICAL LOGARITHMIC SOBOLEV QUESTION

AGENT LANE 14

## STATUS

This packet gives a candidate counterexample to an open question in the source paper [1]. The authors ask whether there is a constant  $a > 0$  such that a logarithmic Sobolev inequality of the form

$$H(e^{-V}\sigma \mid \sigma) \leq \frac{a}{2} \int_{\mathbb{S}^n} \log \left( 1 + \frac{|\nabla_{\mathbb{S}^n} V|^2}{a^2} \right) e^{-V} d\sigma$$

holds on the sphere for all regular enough normalized potentials  $V$ , where  $\int e^{-V} d\sigma = 1$ . The answer is negative in every dimension  $n \geq 1$ .

## SOURCE TARGET

The source asks the question in the introduction, after deriving conditional and unconditional log-Sobolev type inequalities from reverse Blaschke–Santaló and log-Minkowski inputs:

Since a version of the log-Minkowski conjecture is true in the unconditional case [Sar15], we show in Theorem 5.10 that (21) is true with the constant  $c = 4^{n+1}/(n+1)!$  when  $C_1, C_2$  are assumed to be unconditional. In this unconditional setting, contrary to what happens in the Gaussian case, Inequality (21) admits equality cases which are given by couples of unconditional convex bodies  $(C, C^\circ)$  minimizing the volume product and properly normalized to be of volume 1. Without assuming the log-Minkowski conjecture, some weaker statement remains valid, see Theorem 5.5. Contrary to the Gaussian case, we do not know whether the underlying log-Sobolev inequality

$$H(e^{-V} dx \mid \sigma) \leq \frac{a}{2} \int \log \left( 1 + \frac{|\nabla_{\mathbb{S}^n} V|^2}{a^2} \right) e^{-V} d\sigma$$

holds true for all regular enough potentials  $V$ , and some constant  $a > 0$ . We refer to Section 5 for additional remarks and open questions about these improved log-Sobolev inequalities on the sphere.

## PROOF INTUITION

There is almost no room for the constant. If the inequality is linearized at the uniform measure in the first spherical harmonic direction, the ordinary spectral gap of  $\mathbb{S}^n$  forces  $a \leq n$ . On the other hand, highly concentrated von Mises–Fisher densities

$$d\eta_t(\theta) = Z_t^{-1} e^{t\theta_1} d\sigma(\theta)$$

have entropy asymptotic to  $(n/2) \log t$ . The right-hand side has leading term  $(a/2) \log t$ , so all  $a < n$  fail. The critical value  $a = n$  still fails: the leading logarithms cancel, but the next constant term is positive. Thus no  $a > 0$  can work.

## COUNTEREXAMPLE

**Theorem 1.** *Let  $n \geq 1$  and let  $\sigma$  be the normalized surface measure on  $\mathbb{S}^n$ . There is no constant  $a > 0$  such that, for every smooth  $V$  with  $\int_{\mathbb{S}^n} e^{-V} d\sigma = 1$ ,*

$$(1) \quad H(e^{-V} \sigma \mid \sigma) \leq \frac{a}{2} \int_{\mathbb{S}^n} \log \left( 1 + \frac{|\nabla_{\mathbb{S}^n} V|^2}{a^2} \right) e^{-V} d\sigma.$$

*Proof.* Suppose first that (1) holds for some  $a > 0$ . Let  $u(\theta) = \theta_1$  be the first coordinate on  $\mathbb{S}^n$ , and put

$$d\eta_\varepsilon = Z_\varepsilon^{-1} e^{\varepsilon u} d\sigma = e^{-V_\varepsilon} d\sigma.$$

As  $\varepsilon \rightarrow 0$ ,

$$H(\eta_\varepsilon \mid \sigma) = \frac{\varepsilon^2}{2} \int u^2 d\sigma + o(\varepsilon^2),$$

while

$$\frac{a}{2} \int \log \left( 1 + \frac{|\nabla_{\mathbb{S}^n} V_\varepsilon|^2}{a^2} \right) d\eta_\varepsilon = \frac{\varepsilon^2}{2a} \int |\nabla_{\mathbb{S}^n} u|^2 d\sigma + o(\varepsilon^2).$$

Since  $u$  is a first spherical harmonic,

$$\int_{\mathbb{S}^n} |\nabla_{\mathbb{S}^n} u|^2 d\sigma = n \int_{\mathbb{S}^n} u^2 d\sigma.$$

Therefore (1) can hold only if  $a \leq n$ .

It remains to rule out  $a \leq n$ . For  $t > 0$ , set

$$d\eta_t(\theta) = Z_t^{-1} e^{t\theta_1} d\sigma(\theta), \quad V_t(\theta) = -t\theta_1 + \log Z_t.$$

Let  $U = \theta_1$  under  $\eta_t$ . Then

$$H(\eta_t \mid \sigma) = t \mathbb{E}_t U - \log Z_t$$

and

$$|\nabla_{\mathbb{S}^n} V_t|^2 = t^2(1 - U^2).$$

The coordinate  $U$  under  $\sigma$  has density

$$q_n(u) = c_n(1 - u^2)^{n/2-1}, \quad c_n = \frac{\Gamma((n+1)/2)}{\sqrt{\pi} \Gamma(n/2)}.$$

Laplace's method at the endpoint  $u = 1$  gives, with

$$C_n = 2^{(n-2)/2} \frac{\Gamma((n+1)/2)}{\sqrt{\pi}},$$

the asymptotics

$$Z_t = C_n e^t t^{-n/2} (1 + o(1)), \quad \mathbb{E}_t(1 - U) = \frac{n}{2t} + o(t^{-1}),$$

and

$$\mathbb{E}_t \log(1 - U^2) = -\log t + \log 2 + \psi(n/2) + o(1),$$

where  $\psi = \Gamma'/\Gamma$ . Consequently

$$(2) \quad H(\eta_t \mid \sigma) = \frac{n}{2} \log t - \frac{n}{2} - \log C_n + o(1).$$

For the right-hand side of (1),

$$(3) \quad \frac{a}{2} \mathbb{E}_t \log \left( 1 + \frac{t^2(1 - U^2)}{a^2} \right) = \frac{a}{2} \log t + \frac{a}{2} \left( \log \frac{2}{a^2} + \psi(n/2) \right) + o(1).$$

If  $a < n$ , (2) and (3) imply that the difference between the left and right sides of (1) tends to  $+\infty$  along  $\eta_t$ .

The only remaining possible value is  $a = n$ . In that case the difference tends to

$$D_n = -\frac{n}{2} - \log C_n - \frac{n}{2} \left( \log \frac{2}{n^2} + \psi(n/2) \right).$$

Writing  $x = n/2$ , this is

$$D_n = \log 2 + \frac{1}{2} \log \pi - x - \log \Gamma(x + 1/2) + 2x \log x - x\psi(x).$$

For  $x \geq 1/2$ , the standard digamma and Stirling bounds

$$\psi(x) \leq \log x - \frac{1}{2x}, \quad \Gamma(x + 1/2) \leq \sqrt{2\pi} x^x e^{-x}$$

give

$$D_n \geq \frac{1}{2}(1 + \log 2) > 0.$$

Thus the inequality also fails for  $a = n$  and all possible constants have been excluded.  $\square$

#### VERIFICATION AND SCOPE

The counterexample uses smooth potentials  $V_t = -t\theta_1 + \log Z_t$ , so it falls within any reasonable interpretation of “regular enough potentials” in the source question. The proof is symbolic. The script `code/verify_critical_constant.py` prints the critical constants  $D_n$  for  $1 \leq n \leq 20$  as a sanity check; these numerical values are not used in the proof.

This packet answers only the standalone open question about the existence of a universal constant  $a$  in the underlying spherical log-Sobolev inequality. It does not affect the source’s proved conditional or unconditional Blaschke–Santaló consequences.

#### NOVELTY CHECK

The local lightweight indexes were searched for 2307.04393, the paper title, **Blaschke–Santaló**, **spherical log-Sobolev**, **transport entropy**, and **von Mises–Fisher**. Web searches on 2026-07-18 for the exact open-question phrases and close variants found the source arXiv page, the published paper listing, and unrelated spherical sampling/log-Sobolev material, but no later answer to this exact question.

#### HUMAN-REVIEW RECOMMENDATION

Review the endpoint Laplace asymptotics and the final gamma/digamma lower bound for  $D_n$ . These are the only non-formal ingredients. If those checks pass, the answer to the source’s question is a full negative answer.

#### REFERENCES

- [1] M. Fradelizi, N. Gozlan, S. Sadovskiy, and S. Zugmeyer, *Transport-entropy forms of direct and converse Blaschke–Santaló inequalities*, arXiv:2307.04393, 2023.